

ANNOTATED BIBLIOGRAPHY

Spencer, Michael. "Makerspaces, Student Creativity, and Next Generation Jobs."

US Black Engineer and Information Technology, vol. 41, no. 3, 2017, pp.

32-33. JSTOR, www.jstor.org/stable/44419235.

In the article "Makerspaces, Students Creativity, and Next Generation Jobs" author Micheal Spencer explains that Makerspaces are areas where students can learn to create, design, experiment, and learn skills that will help them. These spaces contain equipment to help students keep up and encourage them to problem solve which is needed for new jobs that require more experience, innovation, and creativity. He emphasizes the idea of actually doing hands-on work rather than simply reading or listening. He talks about the need for more STEM education and connects it to the makerspaces including hands on learning. Finally, he believes makerspaces are able to better prepare for future challenges that rely on technology and hands on skills.

This article is a good tool to use for an argument in hands on learning but it has a few issues. A strength of the article is that it connects education to real world skills that show why makerspaces are important to modern learning or future jobs. Another strength is that the idea is friendly and understandable which is supported with examples of what it provides. A weakness of the article is that there is no concrete data that he presents that shows how these makerspaces truly affect job outcomes. Another weakness he ignores is the challenge of funding these spaces since buying and maintaining equipment for a small margin of students. Lastly the article is inspiring and promotes creativity, it fails to address issues the makerspaces could face and doesn't bring up much concrete data.

Kuhail, Mohammad Amin, et al. "'Will I Be Replaced?' Assessing ChatGPT's Effect

on Software Development and Programmer Perceptions of AI Tools." *Science of Computer Programming*, vol. 235, July 2024, p. 103111. *Science Direct*,

<https://doi.org/10.1016/j.scico.2024.103111>.

Kuhail Mohammad's article explains and investigates how ChatGPT 3.5 performs on programming tasks and talks about how programmers feel about using it in their work. For ChatGPT 3.5 it is able to do easy to medium level coding tasks or problems, however for more complex or hard problems it struggles to be accurate or understandable. The study questions programmers trust, perceived productivity, and job security based on the up and coming AI tools. The research shows that programmers do use it and see the benefits, however they still worry that with time they may lose their jobs. In the end it shows that people are convinced that AI can only assist and not fully take over the creativity or true skills of humans.

Valuable insight to how programmers view AI tools in their workspace but has issues with the broad scope. One strength of the article is that it gives actual surveys of programmers and programming tasks rather than guessing theories about the future of AI. Another strength is that from the surveys of programmers it shows the opinions of actual developers and how they feel about it, rather than just data about performance margins. A weakness is it doesn't specify too much about AI other than ChatGPT specifically 3.5 which can limit how the results can apply to the future of AI. Another weakness is that the article lacks context of specific tasks, for example some tasks are different from others and require different types of skills, thought, or creative approach which can not fully reflect all real world professional code. While the article provides a good perspective on trends or results, it would be a lot more useful if it was more broad about the future and also better or deeper real world applications.

Mohammed S. Will AI replace statistical programming jobs. J Manag Res Anal.
2025;12(3):185-189.

https://www.researchgate.net/profile/Shanavaz-Mohammed/publication/395800962_Will_AI_replace_Statistical_Programming/links/68d43f1dd221a404b2a2581f/Will-AI-replace-Statistical-Programming.pdf

In the article "Will AI replace Statistical programming jobs" Mohammed explains that AI is starting to automate many basic tasks in statistical programming such as data cleaning, basic modeling, and code generation. The argument for why these tasks are taken by AI is that they are simple and are easy for AI and more efficient than people since they follow a repeatable pattern. On the other hand he argues that AI cannot fully replace human programming since they are needed for complex decision making and understanding real world problems. He says humans must oversee AI tools to make sure they are ethical and accurate. He believes AI will change statistical programming jobs, but not completely erase them.

This article is a good choice to provide a clear overview of how AI affects statistical programming, but it has some considerations. A strength of the article is that it shows what tasks AI can automate and what tasks still require human input. Another strength is that it gives examples of how AI is already being used in today's data work. A weakness of the article is that it shows little amounts of real studies done to prove the predictions being made. Another main weakness is that like a lot of these articles they assume humans will be able to adapt easily to the growth of AI, but don't discuss the job loss or other losses assumed with AI taking over. Overall the article is a good source for a general argument or studies but it would be stronger with a deeper thinking on human matters and even evidence.

Morgan Stanley Alphawise. "How AI Coding Is Creating Jobs." *Morgan Stanley*, 29 Oct. 2025,
www.morganstanley.com/insights/articles/ai-software-development-industry-growth?

In the article "How AI Coding is Creating Jobs" by Morgan Stanley explains how AI tools are simply changing the way software development is done, but not in a way that eliminates all human jobs. Like the other articles it argues that AI takes over repetitive tasks which allows developers to focus on more difficult problems that require complex thought. The article

highlights that as technology evolves so will the need for software, so the demand for software jobs will increase which will create more total jobs. The article argues that new roles will be created as a result and therefore become mainstream and important. This article argues that the introduction of AI will expand the software industry not shrink it.

This article has a good optimistic view of how AI will impact the job market but it ignores some of the issues that come up within the topic. A strength of the article is that it shows statistics and data to show that jobs are being created or expanded. Another strength of the article is that it gives actual examples of how AI supports programmers instead of simply stating their being replaced. A weakness of the article is that it only really looks at the positives and not the negatives such as job loss or risks associated with the introduction of AI. Another weakness is that it only looks at big companies or corporations instead of small indie studios or other small businesses that may be more affected. Overall the article is good for information about big companies but it fails to be balanced in its perspective against other smaller businesses.

Gordon, Rachel. "Can AI Really Code? Study Maps the Roadblocks to Autonomous

Software Engineering." *MIT News*, MIT, 16 July 2025,

<https://news.mit.edu/2025/can-ai-really-code-study-maps-roadblocks-to-autonomous-software-engineering-0716>

The article "Can AI Really Code? Study Maps the Roadblocks to Autonomous Software Engineering" Author Rachel Gordon discusses the limits of AI and its coding. She talks about how AI struggles with complex logic and truly understanding or learning a project. Another thing worth noting is that the AI struggles or breaks even when it requires deeper or much longer thinking. She uses the same argument that AI is useful for small repetitive tasks

that can be done without deep thinking. The same argument is used to show that humans are still needed for oversight and guiding AI outputs. Lastly, the article concludes that fully autonomous coding is still not realistic and is far from reality.

This article provides a strong insight in the technicalities of AI and its limits but it doesn't have the broader scope that's hoped for. One strength of the article is that it uses real research from her school to back up her argument. Another strength of the article is that it breaks down the limits of AI coding systems which helps with understanding why AI cannot work independently as of right now. A weakness of the article is that it only focuses on the limitations of today and doesn't consider or explore how future systems or advancements might overcome the challenges seen today. Another weakness of the article is that it doesn't address how the limits might affect workers, businesses, or education which leaves the real consequences unclear. Overall the article is a good source for technical detail it lacks the thought of the future which would be an important thing to consider.

Chong, Ng S.T. "AI vs Complex Code: Why Software Engineers Still Matter in

2025." *UNU*, United Nations University, 28 Aug. 2025,

<https://c3.unu.edu/blog/ai-vs-complex-code-why-software-engineers-still-matter-in-2025?>

In the article "AI vs Complex Code: Why Software Engineers Still Matter in 2025" the University conducted a software project using a few advanced AI coding tools in order to see how effective they were in building a complex chart management interface. At first the AI was able to perform on simple and repetitive tasks, however, as it was applied for harder multi layering logic or reasoning tasks, it began to fail more. They explain that AI could not manage long term planning, important structural decisions, or context switching, which then forced real programmers to intervene to fix and redesign the structure of code. The AI couldn't understand the full purpose of the project or each task performed, which meant that the process still

required human input for unpredictable issues. Overall the article shows that while AI can help with the basics, we still need real input from programmers for the most essential parts of development.

This article is valuable since it provides good insight to the limits of AI in a clear way, but still has its limitations. One strength that the article has is that instead of using predictions based on a guess, it uses real experiences with the AI which makes it a lot more credible. Another strength is that it shows specifically where and why parts failed and shows the limits when logic layers, and long term planning are involved. A weakness is that it's based on one coding project which isn't a great indicator of other real world scenarios. Another weakness is that it doesn't show how much human involvement is needed in order to fix it, but rather it shows the steps of how to fix it instead. Overall the article is a great source for how human programmers still matter in coding, but it leaves out broader factors and other research.

Brodsky, Sascha. "AI Can Write Code, but Can It Beat Software Engineers?" *IBM News*, IBM, 2025, www.ibm.com/think/news/ai-write-code-can-beat-software-engineers.

In the article "AI Can Write Code, but Can It Beat Software Engineers?" Writer Sascha Brodsky explains where AI succeeds or fails in coding work. The article reviews research that was done by MIT about how AI models can only handle short and simple tasks rather than longer and more complex tasks. The researchers found that it can write pieces of code quickly but struggle to understand how the code connects to a large real world project. Because of these issues human programmers are still needed to make important decisions and fix issues that AI cannot do on its own. Overall The article says that AI is able to work faster than humans but they are still needed for the more thoughtful and important decisions of software development.

This article offers a good explanation of the limits of AI and contrasts it with another source, but it leaves out broader factors. A strength of the article is that it reviews information from MIT and combines it with their knowledge which makes the article very credible. Another strength is that it explains why AI struggles in general but especially with larger projects that require lots of planning and big decisions. A weakness of the article is that the article mostly focuses on the limits of AI rather than providing examples of where it succeeds. Another weakness is that it doesn't talk about how AI might develop in the future which can leave out important factors to consider. Overall the article is a helpful source for why AI is good at assisting in coding, but humans are still needed for reasoning and judgement.

Friedman, Itamar, Co-founder & CEO. "2025 State of AI Code Quality." Qodo, 2025,
www.qodo.ai/reports/state-of-ai-code-quality/.

In the article "2025 State of AI Code Quality" By Qodo explains how programmers view and experience AI tools in current software development. The article shows that many programmers believe AI is helpful and improves productivity since it can do basic tasks faster and with less stress. It also explains that many programmers believe AI can sometimes actually improve code quality, however, there are still mistakes that are made that need to be corrected manually. Since AI cannot always understand real world requirements, programmers will still need to be responsible for the accuracy and safety of their code and fix it accordingly. Lastly the article still talks about how human oversight will still be necessary in order for AI generated code to be credible or dependable in actual projects.

This article provides an amazing amount of evidence about code quality, but it has its issues. A strength of the article is that it uses real statistics and data from programmers which shows that it backs up its arguments. Another strength is that it discusses the limits and the benefits of using AI to do coding tasks which gives a credible and balanced viewpoint. A weakness of the article is that it only speaks on what has happened or present tools which doesn't take into account future developments of AI tools which might change situations or outcomes later in the future. Another weakness is that it only gives statistics and doesn't really go into specific examples of when or why errors occurred and caused complications in real projects. Overall the article is a great source to find statistics and data on the current use of AI tools in coding, but it also shows how humans are still needed for quality and safety.

EXTRA SOURCES

- <https://www.devopsdigest.com/ai-wont-replace-developers-it-will-make-the-best-ones-in-dispensable>
- <https://www.cio.com/article/4080210/the-engineering-imperative-why-ai-wont-replace-our-best-developers.html?>

